Shajeenth Shanmugarajah

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Mechatronics Engineering — *Sep. 2026 – Present*

Victoria Park Collegiate Institute — North York, ON
IB Diploma, OSSD Diploma, High School Student — *Sep. 2022 – July. 2026*

Projects

ROS 2 LiDAR Navigation Simulation | *ROS 2, C++, Foxglove*

- Created four ROS 2 nodes to control autonomous navigation of a robot when clicking on a position on a grid using LiDAR data
- Implemented occupancy costmap creation for obstacle representation, inflating obstacle cells and creating zero-valued cells for navigation
- Implemented A* for path creation and pure pursuit navigation logic

FreeRTOS Gripper | *ESP-IDF, C++, ESP32, PWM, Solidworks*

- Developed a Wi-Fi-controlled gripper using FreeRTOS and ESP-IDF
- Built a web-based control panel hosted on the ESP32, communication with Wi-fi
- 3D printed and designed the gripper mechanism in Solidworks, implementing servo control with the printed parts

Makeshift Controller | *C++, ESP32, BLE*

- Designed a custom Bluetooth HID game controlled using an ESP32, integrating analog joysticks, multiple button inputs, and haptic feedback
- Implemented haptic feedback and mapped it to specific button inputs
- Debugged and tuned input mapping so that it works with various Steam games, such as Hollow Knight and Hades

Experience

Victoria Park Robotics Programming Lead and Member — Sep. 2023 – Mar. 2026
Victoria Park Collegiate Institute — *North York, ON*

- Developed a centralized static Java class for managing motor and servo IDs improving hardware configuration consistency in order to boost efficiency in programming
- Tested and tuned motor speed and position to achieve accurate and repeatable linear slide positioning using encoder feedback
- Worked amongst a team using one shared Github repository
- Competed in the First Tech Challenge and won an Innovate award

Victoria Park Coding Club Executive — Sep. 2024 – Feb. 2026
Victoria Park Collegiate Institute — *North York, ON*

- Taught advanced competitive programming concepts in Python, Java, and C++ to students in order to succeed in the CCC and other competitive programming contests
- Planned trips to other schools and events in a team of 5 students in order to compete in practice competitive programming competitions

Technical Skills

Languages: C/C++, Java, Python, Typescript, JavaScript, HTML/CSS
Frameworks: ESP-IDF, FreeRTOS, ROS 2, React, Node.js
Developer Tools: Git, Docker, CMake, VS Code, Visual Studio, PyCharm, Solidworks, Arduino IDE
Libraries: OpenCV, NumPy, Tensorflow, BleGamepad